SI 140A.02 Probability & Statistics for EECS, Fall 2026
Homework 6

Name:
Student ID:

Due on Nov. 1, 2026, 23:59 UTC+8

Read all the instructions below carefully before you start working on the assignment, and before you make a submission.

- You are required to submit your homework in PDF format generated by **LaTeX**, otherwise you will get no points of this homework.
- You are required to write down all the major steps towards making your conclusions; otherwise you may obtain limited points of the problem.
- Write your homework in English; otherwise you will get no points of this homework.
- Any form of plagiarism will lead to 0 point of this homework.

Problem 1 (20 points)

Let $n \geq 3$, and let $X_1, \ldots, X_n$ be binary random variables with $\Pr(X_i = 1) = p_i \in (0, 1)$. Suppose that every collection of $n - 1$ variables is mutually independent. No symmetry of the joint distribution is assumed.

(a) Determine all possible joint PMFs and the exact admissible range of their free parameter. Prove both necessity and sufficiency.

(b) Suppose additionally that $n = 4$, $p_i = 1/2$ for all i, and all 16 joint outcomes have strictly positive probability. Determine the exact range of $\Pr(X_1 + X_2 + X_3 + X_4 = 2)$. Does the value $3/8$ force mutual independence?

Problem 2 (20 points)

A population contains m red balls and m blue balls. A uniformly chosen sample of size n is drawn without replacement. Let $X_{m,n}$ be the number of red balls, and let $Y_n \sim \mathrm{Bin}(n, 1/2)$.

(a) For fixed positive integer n and fixed $k \in \{0, \ldots, n\}$, give the PMF of $X_{m,n}$ and evaluate

$$\lim_{m\to\infty} m\left(\frac{\Pr(X_{m,n} = k)}{\Pr(Y_n = k)} - 1\right).$$

Do not merely quote convergence of the hypergeometric distribution.

(b) Set $n = 2m - 1$ and $A_m = \{m - 1, m\}$. Evaluate

$$\lim_{m\to\infty} \left[\Pr(X_{m,2m-1} \in A_m) - \Pr(Y_{2m-1} \in A_m)\right].$$

Does a growing population with a fixed red proportion alone justify the binomial approximation? Prove the required limit without the central limit theorem or Stirling's formula.

Problem 3 (20 points)

Let X take values in $\{1, 2, 3, 4\}$, with zero probabilities allowed. Suppose

$$\Pr(X \in \{1, 2\}) = \frac{2}{3}, \qquad \Pr(X \in \{2, 3\}) = \frac{1}{2}.$$

Entropy is measured in bits, with $0 \log_2 0 = 0$.

(a) Determine the maximum and minimum of $H(X)$ and all PMFs attaining them. Prove that the extrema are global.

(b) If all four probabilities must be strictly positive, determine the exact set of possible entropy values and state which boundaries are attained.

Problem 4 (20 points)

Let $N \geq 2$ be an integer. A random variable X takes values in $\{0, 1, \ldots, N\}$ and satisfies $\mathbb{E}[X] = \mu$, where $0 < \mu < N$. No other distributional assumptions are imposed.

(a) Determine the minimum and maximum possible variance and characterize all equality cases. Cover both integer and noninteger values of μ.

(b) Prove that every value between these extrema is attainable, and provide an explicit PMF construction.

Problem 5 (20 points)

Consider independent Bernoulli trials with success probability $p \in (0, 1)$. Fix an integer $r \geq 2$, and let T_r be the trial number of the rth success, including the successful trial.
Fix an integer $m \geq r - 1$. An observer is told only that $T_r > m$, not how many successes occurred during the first m trials. On this event, let $R_m = T_r - m$.

(a) Find the complete conditional PMF of R_m given $T_r > m$. Express it also as a mixture of waiting-time distributions, specifying the mixture weights.

(b) Can this conditional distribution be a first-success distribution with any parameter? For each fixed integer $k \geq 1$, evaluate

$$\lim_{m \to \infty} \Pr(R_m = k \mid T_r > m)$$

and identify the limiting distribution.